

JINING, China

WMO: 534800

Lat: 41.0295N

Lon: 113.0704E

Elev: 4646

StdP: 12.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-5.3	-24.2	1.7	1.1	-20.4	2.1	3.0	23.4	8.8	20.5	10.0	2.8	290	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	84.9	60.6	82.2	59.8	79.8	59.0	66.8	75.8	65.1	74.3	63.4	73.1	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	106.7	70.5	62.1	99.4	69.5	60.1	92.3	68.5	34.2	75.5	32.8	74.5	31.4	72.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	17.8	15.0	DB	-14.5	88.8	3.2	2.8	-16.8	90.9	-18.6	92.5	-20.4	94.1	-22.7	96.2	(4)
(5)			WB	-15.1	69.2	3.1	2.4	-17.4	70.9	-19.2	72.3	-20.9	73.6	-23.2	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	10.4	17.7	30.3	44.9	56.0	64.5	68.9	65.7	56.1	42.9	27.3	14.1		
	DBStd	21.61	7.85	8.99	9.37	9.29	7.42	5.40	4.30	4.59	6.51	7.78	8.97	7.79		
	HDD50	5032	1227	903	611	203	29	1	0	22	241	682	1113			
	HDD65	8780	1692	1323	1074	603	291	70	12	46	274	685	1132	1578		
	CDD50	2003	0	0	1	51	215	437	586	488	204	21	0	0		
	CDD65	278	0	0	0	1	11	57	134	69	6	0	0	0		
	CDH74	2521	0	0	0	21	184	596	1119	529	72	0	0	0		
	CDH80	502	0	0	0	2	25	118	270	77	10	0	0	0		
	WSAvg	5.4	4.9	5.6	6.5	7.2	6.8	5.3	4.2	3.9	4.4	5.4	5.5	5.5		
	PrecAvg	14.2	0.1	0.1	0.3	0.7	0.9	2.1	3.5	4.0	1.6	0.6	0.2	0.1		
(15) Precipitation	PrecMax	21.1	0.4	0.8	1.0	3.1	2.3	5.4	7.0	8.0	4.1	1.6	0.6	0.3		
	PrecMin	6.6	0.0	0.0	0.0	0.0	0.1	0.2	1.0	0.7	0.1	0.0	0.0	0.0		
	PrecStd	3.7	0.1	0.2	0.2	0.6	0.5	1.1	1.6	2.1	0.9	0.5	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	46.1	62.1	76.8	83.6	87.1	88.8	85.8	80.8	70.1	54.6	37.8		
		MCWB	27.1	33.4	41.8	49.8	55.0	59.0	63.1	61.5	56.9	51.0	39.7	28.4		
	2%	DB	30.7	40.6	55.6	71.1	78.9	83.4	85.6	82.0	75.6	64.9	48.8	32.9		
		MCWB	23.3	30.1	38.7	47.1	52.6	57.8	62.3	61.6	55.2	47.9	36.6	25.6		
	5%	DB	26.8	36.3	50.6	66.2	75.3	80.5	82.9	79.4	72.2	60.8	44.6	29.3		
		MCWB	20.7	27.3	35.7	44.8	50.6	56.5	61.5	60.6	54.1	45.2	34.2	23.2		
	10%	DB	23.4	32.7	46.2	61.5	71.6	77.4	80.2	76.9	69.0	56.9	40.4	26.2		
		MCWB	18.3	25.5	33.4	42.6	49.2	55.9	61.1	59.9	52.8	43.8	32.1	21.1		
	0.4%	WB	27.8	34.4	43.2	52.9	58.0	65.1	70.0	68.6	63.3	53.1	41.7	29.4		
		MCDB	35.3	43.9	57.9	69.8	74.7	75.0	78.9	77.3	71.6	65.3	50.0	36.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.7	31.0	39.4	48.9	55.1	62.5	67.5	66.3	59.4	50.2	37.8	26.2		
		MCDB	30.1	38.6	53.0	65.4	72.1	74.0	76.9	74.9	68.6	61.6	47.4	31.9		
	5%	WB	21.0	28.2	36.6	46.4	53.2	60.8	65.9	64.7	56.8	47.6	35.2	23.6		
		MCDB	26.6	35.7	49.4	62.5	68.9	72.8	75.4	73.4	66.7	57.2	43.0	29.0		
	10%	WB	18.5	25.7	34.0	44.0	51.3	59.1	64.2	62.9	54.8	45.3	32.7	21.1		
		MCDB	23.3	32.7	45.1	58.7	66.8	71.3	73.8	71.9	65.3	54.5	40.1	26.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	20.5	21.3	21.9	22.0	20.4	18.6	19.0	20.3	20.0	18.3	17.7		
		MCDBR	22.9	24.5	26.4	28.3	27.9	25.7	23.7	23.1	24.9	25.6	23.0	21.1		
		MCWBR	17.4	16.8	15.7	14.3	12.3	9.7	7.9	8.6	10.6	12.8	14.0	15.6		
		MCDBR	22.1	22.5	25.2	24.8	23.0	19.4	17.4	17.1	19.9	21.8	21.2	20.5		
	5% WB	MCWBR	17.0	15.9	15.9	13.5	11.4	8.5	8.1	8.0	10.2	12.4	13.5	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.344	0.380	0.437	0.480	0.535	0.505	0.496	0.449	0.387	0.319	0.295		
	taud		2.266	2.109	2.049	1.965	1.899	1.829	1.963	1.959	2.008	2.112	2.239	2.260		
	Ebn at Noon		273	269	272	264	255	241	247	245	247	249	256	259		
	Edh at Noon		29	40	48	56	61	66	57	56	49	39	30	27		
(44) All-Sky Solar Radiation	RadAvg		807	1108	1551	1873	2084	1933	1879	1761	1492	1200	849	699		
	RadStd		73	94	126	117	135	155	91	111	84	66	91	58		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-4.23	N/A	-0.97	-1.48	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.76	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page